

SKUNKWORKS ACADEMY

# CESE323 ERC1.0 - Cloud Security Envisioning & Strategy

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*Student Lab Exercises Guide*

*Microsoft Defender for Cloud, CSPM, Workload Protection, AI Security, and  
Operational Review*

|                  |                                        |
|------------------|----------------------------------------|
| Document version | ERC1.0                                 |
| Audience         | Students / delegates                   |
| Format           | Instructor-led or self-paced lab guide |
| Date             | 25 May 2026                            |

*Student Edition - no instructor solution notes or answer key included*

## Document Control

| Item                    | Value                                                                                        |
|-------------------------|----------------------------------------------------------------------------------------------|
| Course code             | CESE323                                                                                      |
| Release                 | ERC1.0                                                                                       |
| Document title          | Student Lab Exercises Guide                                                                  |
| Intended audience       | Students attending or completing the Cloud Security Envisioning & Strategy course            |
| Security classification | Training material - student use                                                              |
| Assumed lab platform    | Microsoft Azure, Microsoft Defender for Cloud, Microsoft Defender portal, Microsoft Entra ID |

**Student edition note:** This guide gives lab objectives, tasks, evidence requirements, worksheets, and safety instructions. It intentionally excludes instructor hints, suggested answers, and scoring notes.

## Guide Map

Use this guide during the hands-on portion of the CESE323 course. Your instructor will confirm which workload modules are included. For the one-day version, complete the required core labs and only the two workload labs selected for your class.

| Lab | Exercise                                       | Recommended time | Required / Optional                | Primary output                             |
|-----|------------------------------------------------|------------------|------------------------------------|--------------------------------------------|
| 00  | Access and Environment Validation              | 15 min           | Required                           | Access confirmation and blocker log        |
| 01  | Security Goals, Scope, and Risk Baseline       | 25 min           | Required                           | Risk baseline worksheet                    |
| 02  | Defender for Cloud Posture Baseline            | 35 min           | Required                           | Secure Score and recommendations evidence  |
| 03  | Regulatory Compliance and Policy Alignment     | 30 min           | Required                           | Compliance gap worksheet                   |
| 04  | Workload Protection Plan Review and Selection  | 25 min           | Required                           | Selected workload plan matrix              |
| 05A | Defender for Storage                           | 40 min           | Optional - selected track          | Storage protection evidence                |
| 05B | Defender for App Service                       | 40 min           | Optional - selected track          | App Service posture evidence               |
| 05C | Defender for Containers                        | 45 min           | Optional - selected track          | Container / AKS posture evidence           |
| 05D | Defender for AI Services and Microsoft Foundry | 45 min           | Optional - selected track          | AI workload security evidence              |
| 06  | Threat Mapping and Remediation Planning        | 35 min           | Required                           | Prioritized remediation plan               |
| 07  | Operational Review Capstone                    | 45 min           | Required                           | Executive-style operational review summary |
| 08  | Cleanup and Cost Control                       | 20 min           | Required if resources were created | Cleanup confirmation                       |

## Lab Evidence Rules

- Keep screenshots focused on the task outcome. Mask subscription IDs, tenant IDs, object IDs, user names, IP addresses, and any sensitive operational data where required.
- Do not upload real customer, personal, financial, HR, legal, regulated, confidential, or proprietary data into any lab workload.

- Do not enable paid Defender plans or deploy agents/extensions in a live subscription unless the instructor confirms written approval and scope.
- Use named student accounts only. Do not share credentials or use generic administrator accounts for evidence.
- Record blockers immediately. A permission or portal access blocker is a valid lab finding when documented clearly.

## Minimum Lab Readiness Checklist

| Readiness item                                                  | Student confirmation      |
|-----------------------------------------------------------------|---------------------------|
| Microsoft Learn account available                               | Yes / No                  |
| Azure portal opens successfully                                 | Yes / No                  |
| Correct Microsoft Entra tenant selected                         | Yes / No                  |
| Assigned Azure subscription or resource group visible           | Yes / No                  |
| Microsoft Defender for Cloud opens successfully                 | Yes / No                  |
| Microsoft Defender portal opens successfully where required     | Yes / No                  |
| MFA configured and tested                                       | Yes / No                  |
| Browser and network access working                              | Yes / No                  |
| Approval confirmed for any paid plan or live environment change | Yes / No / Not applicable |

## Lab 00 - Access and Environment Validation

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 15 minutes                                                  |
| Delivery mode  | Individual or paired                                        |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Confirm that you can access the portals and lab subscription required for the course.
- Verify that Microsoft Defender for Cloud baseline pages load correctly.
- Identify permission, network, MFA, or tenant selection blockers before configuration labs begin.

### Scenario

You are preparing to participate in a cloud security workshop. Before reviewing posture or enabling workload protection, you must prove that your account, browser, tenant, subscription, and Defender views are ready.

### Before you start

- A named student account with MFA enabled.
- Access to the Azure portal and the correct Microsoft Entra tenant.
- At minimum Reader access to the subscription or resource group used for observation labs.
- Contributor, Security Admin, or Owner only where explicitly required for the selected lab track.

**Safety rule:** Stop if the subscription or tenant is not the one assigned to you. Do not make changes in production or customer subscriptions unless your instructor confirms the scope.

### Student tasks

1. Sign in to the Azure portal and confirm your account name, tenant, and subscription context.

2. Open Microsoft Defender for Cloud and confirm that Overview, Environment settings, Recommendations, Inventory, Secure Score, and Regulatory compliance can load.
3. Open the Microsoft Defender portal if your class includes Defender XDR incident or alert review.
4. Confirm whether you have Reader, Contributor, Security Admin, or Owner access for the assigned scope.
5. Create a blocker note for any missing access, unexpected tenant, unavailable portal page, network restriction, or MFA issue.

### Evidence to submit

- Screenshot showing Defender for Cloud successfully loaded, with sensitive identifiers masked where required.
- Completed readiness checklist.
- Blocker log entry if any required service or permission is unavailable.

### Student worksheet

| Check | Expected result | Actual result | Blocker / action required |
|-------|-----------------|---------------|---------------------------|
|       |                 |               |                           |
|       |                 |               |                           |
|       |                 |               |                           |
|       |                 |               |                           |
|       |                 |               |                           |

### Reflection checkpoint

- What was the most important access dependency for this lab?
- What permission would you request if you could not view posture data?

## Lab 01 - Security Goals, Scope, and Risk Baseline

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 25 minutes                                                  |
| Delivery mode  | Individual, paired, or team-based                           |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Translate business and operational security goals into cloud security review priorities.
- Define the lab scope before reviewing posture, recommendations, workload plans, or live environment settings.
- Create a risk baseline that will be refined during later Defender for Cloud exercises.

### Scenario

A customer wants a practical cloud security review that moves beyond theory. Your first task is to define the business priorities, in-scope cloud assets, and risk assumptions that will guide the hands-on review.

### Before you start

- Course scenario or customer environment description supplied by the instructor.
- Access to Azure and Defender for Cloud is useful but not required for this planning exercise.

**Safety rule:** Do not record confidential customer details in the worksheet. Use generalized descriptions where required.

### Student tasks

- Identify the organization or scenario being reviewed and describe the in-scope environment in one paragraph.
- List at least three security goals, such as reducing over-privileged identities, improving Secure Score, hardening storage, reducing external exposure, or protecting AI workloads.
- List the cloud workloads that are in scope: subscriptions, resource groups, VMs, storage, App Service, containers, AKS, databases, AI services, or Microsoft Foundry resources.
- Create five initial risk statements using this format: Because of [condition], [asset/workload] may be exposed to [threat/impact].
- Map each risk statement to a Defender for Cloud or Microsoft security capability that could help detect, prioritize, or reduce the risk.

### Evidence to submit

- Completed risk baseline worksheet.
- List of in-scope and out-of-scope assets.
- Three to five measurable security goals for the lab environment.

### Student worksheet

| Risk ID | Asset / scope | Risk statement | Likely impact | Relevant Microsoft security capability |
|---------|---------------|----------------|---------------|----------------------------------------|
|         |               |                |               |                                        |
|         |               |                |               |                                        |
|         |               |                |               |                                        |
|         |               |                |               |                                        |
|         |               |                |               |                                        |

### Reflection checkpoint

- Which risk is most likely to create business impact?
- Which risk can be measured using Defender for Cloud posture data?

## Lab 02 - Defender for Cloud Posture Baseline

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 35 minutes                                                  |
| Delivery mode  | Individual or paired                                        |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Use Defender for Cloud to review Secure Score, recommendations, affected resources, and security posture signals.
- Identify misconfigurations, vulnerable resources, and over-privileged or risky configuration patterns.
- Prioritize posture findings using risk, affected asset criticality, and implementation effort.

### Scenario

You are acting as a cloud security analyst. Your goal is to establish a posture baseline for the assigned subscription or resource group and identify the first set of findings that should be remediated.

### Before you start

- Defender for Cloud enabled in the lab subscription or viewable in the assigned scope.
- Reader or higher permissions for posture review.
- Secure Score, Recommendations, Inventory, and Regulatory compliance pages visible.

**Safety rule:** Do not apply remediation unless your instructor explicitly authorizes changes in the assigned lab resource group.

### Student tasks

11. Open Defender for Cloud and record the current Secure Score or posture summary shown for your assigned scope.
12. Open Recommendations and filter or sort findings by severity, risk, or affected resources as available.
13. Select three findings that are relevant to the course objectives: misconfiguration, identity risk, workload exposure, weak hardening, missing monitoring, or regulatory gap.
14. For each finding, record the affected resource type, severity or risk signal, suggested remediation, and likely operational impact.
15. Open Inventory or Cloud Security Explorer if available and identify one affected resource that should be investigated further.
16. If attack path analysis or exposure management is available, note whether any attack path, choke point, or exposure graph is visible for your selected findings.

### Evidence to submit

- Screenshot of Secure Score or posture overview.
- Screenshot or export of selected recommendations, with sensitive identifiers masked if needed.
- Completed posture baseline worksheet with three prioritized findings.

### Student worksheet

| Finding ID | Resource / type | Severity / risk | Why it matters | Recommended action | Priority |
|------------|-----------------|-----------------|----------------|--------------------|----------|
|            |                 |                 |                |                    |          |
|            |                 |                 |                |                    |          |
|            |                 |                 |                |                    |          |
|            |                 |                 |                |                    |          |

### Reflection checkpoint

- Which finding had the strongest business risk?
- Which finding would be easiest to remediate without service disruption?

## Lab 03 - Regulatory Compliance and Policy Alignment

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 30 minutes                                                  |
| Delivery mode  | Individual or paired                                        |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Review compliance posture in Defender for Cloud.
- Relate cloud technical findings to governance, risk, and compliance requirements.

- Identify control gaps that require remediation evidence, owner assignment, or exception handling.

### Scenario

The customer needs to understand how technical posture findings translate into compliance status. You will review the compliance dashboard and identify the controls that require evidence or remediation.

### Before you start

- Defender for Cloud Regulatory compliance dashboard visible.
- At minimum Reader access to the target subscription or resource group.
- Awareness of the compliance framework or benchmark used in the lab scenario.

**Safety rule:** Do not assign new standards or policies in a customer or production subscription unless explicitly authorized.

### Student tasks

- Open Defender for Cloud and navigate to Regulatory compliance.
- Identify the standards, benchmarks, or initiatives currently assigned to the environment.
- Choose one standard or benchmark and identify at least three controls that are failed, partially satisfied, or require review.
- For each selected control, identify the associated resource or recommendation where available.
- Classify each control gap as technical remediation, policy exception, evidence collection, or ownership issue.
- Describe what evidence would be required to demonstrate improvement or closure.

### Evidence to submit

- Screenshot of the regulatory compliance dashboard or selected standard.
- Completed compliance gap worksheet.
- One short paragraph explaining how the compliance view supports governance discussions.

### Student worksheet

| Control / requirement | Current status | Related resource / recommendation | Gap type | Evidence needed |
|-----------------------|----------------|-----------------------------------|----------|-----------------|
|                       |                |                                   |          |                 |
|                       |                |                                   |          |                 |
|                       |                |                                   |          |                 |
|                       |                |                                   |          |                 |

### Reflection checkpoint

- Which control gap would you escalate first?
- Which control gap requires evidence rather than technical change?

## Lab 04 - Workload Protection Plan Review and Selection

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 25 minutes                                                  |
| Delivery mode  | Individual or paired                                        |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Identify the Defender for Cloud workload protection plans relevant to the selected labs.

- Compare storage, container, App Service, and AI workload protection scope.
- Document prerequisites, cost approval needs, and change approval requirements before enabling plans.

### Scenario

The syllabus includes a workload protection selection where two out of four workload modules are selected. Your task is to review available plans and document which workload protections are relevant to your lab or customer scenario.

### Before you start

- Defender for Cloud Environment settings visible.
- Coverage workbook visible if available.
- Instructor confirmation of the two selected workload modules for the class.

**Safety rule:** Review plan status first. Do not enable or disable workload protection plans unless the instructor confirms the subscription scope, cost approval, and rollback plan.

### Student tasks

23. Open Defender for Cloud > Environment settings and locate the assigned subscription or management group.
24. Review the available Defender plans and identify current status for Storage, Containers, App Service, AI Services or AI workloads, Servers, SQL, and other relevant workloads.
25. Record whether each selected workload plan is Enabled, Disabled, Not available, or Not in scope.
26. For each selected workload, document the prerequisite resource, required permission, possible cost impact, and rollback or cleanup action.
27. Confirm the two workload labs you will complete and mark the remaining two as optional or observation-only.

### Evidence to submit

- Screenshot of Environment settings or Coverage workbook if permitted.
- Completed workload protection selection matrix.
- Written confirmation of which two workload tracks are selected for your class.

### Student worksheet

| Workload plan | Status | Required resource | Permission needed | Cost / change approval note | Selected? |
|---------------|--------|-------------------|-------------------|-----------------------------|-----------|
|               |        |                   |                   |                             |           |
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|               |        |                   |                   |                             |           |
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|               |        |                   |                   |                             |           |

### Reflection checkpoint

- Which workload protection plan has the highest operational impact?
- Which plan would you not enable without additional approval?

## Lab 05A - Defender for Storage

| Field          | Details                 |
|----------------|-------------------------|
| Estimated time | 40 minutes              |
| Delivery mode  | Optional workload track |



|                |                                                             |
|----------------|-------------------------------------------------------------|
|                |                                                             |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Review storage workload protection posture using Defender for Cloud.
- Identify suspicious access, malware scanning, data exposure, and configuration risks relevant to storage accounts.
- Document safe hardening actions for Azure Storage without using sensitive data.

### Scenario

A storage account is used to hold application files and exported business data. You must review whether the storage workload is protected and identify hardening improvements without exposing real data.

### Before you start

- Azure subscription and Defender for Cloud enabled.
- A lab storage account and blob container assigned to you or created by the instructor.
- Permission to view or configure Defender for Storage for the assigned scope.
- Only benign test files. No customer data, confidential documents, PII, malware, or unauthorized test strings.

**Safety rule:** Use only instructor-approved benign test files. Do not upload real malware or sensitive data. Do not enable malware scanning in a paid plan unless approval is confirmed.

### Student tasks

28. Locate the assigned storage account in the Azure portal and record its resource group, region, redundancy, and public access status where visible.
29. Open Defender for Cloud recommendations related to storage and identify findings for encryption, public access, network restrictions, secure transfer, keys, data protection, or malware scanning.
30. Review Defender for Storage plan settings for the assigned scope. Record whether the plan is enabled and which components are available.
31. If instructed, upload a small benign text file to the lab blob container and verify that no confidential data is included.
32. Review storage-related recommendations, alerts, activity logs, or security insights available in Defender for Cloud.
33. Document at least three hardening actions that would reduce data exposure or suspicious access risk.

### Evidence to submit

- Screenshot of the lab storage account overview or container, with identifiers masked if required.
- Screenshot of storage-related Defender recommendations or plan settings.
- Completed storage workload worksheet.

### Student worksheet

| Storage finding | Risk | Recommended hardening action | Can student apply? | Evidence captured |
|-----------------|------|------------------------------|--------------------|-------------------|
|                 |      |                              |                    |                   |
|                 |      |                              |                    |                   |
|                 |      |                              |                    |                   |
|                 |      |                              |                    |                   |

### Reflection checkpoint

- Which storage risk could expose business data fastest?

- Which hardening action would you apply first in a production environment?

## Lab 05B - Defender for App Service

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 40 minutes                                                  |
| Delivery mode  | Optional workload track                                     |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Review App Service posture and runtime protection coverage.
- Identify common cloud-native application security findings.
- Document safe improvements for diagnostics, TLS, managed identity, network access, and configuration hardening.

### Scenario

A sample web application is hosted on Azure App Service. Your task is to review its security posture and recommend improvements that reduce exposure without disrupting a production workload.

### Before you start

- A lab App Service plan and sample web app assigned to the student or group.
- Defender for Cloud visible for the subscription or resource group.
- Permission to view App Service configuration and recommendations.

**Safety rule:** Only change settings on lab apps assigned to you. Do not modify production web apps.

### Student tasks

34. Locate the assigned App Service and record its resource group, runtime stack, region, and App Service plan tier where visible.
35. Review configuration items such as HTTPS only, minimum TLS version, FTPS state, authentication, managed identity, diagnostic settings, and networking options.
36. Open Defender for Cloud recommendations filtered for App Service or application workloads.
37. Identify at least three application security posture findings or hardening opportunities.
38. If instructed, apply one low-risk hardening setting to the lab app and record the before and after state.
39. Document recommended monitoring or logging improvements for operational review.

### Evidence to submit

- Screenshot of the assigned App Service overview.
- Screenshot of App Service security or configuration setting reviewed.
- Completed App Service posture worksheet.

### Student worksheet

| App Service control | Current state | Risk / concern | Recommended state | Change applied? |
|---------------------|---------------|----------------|-------------------|-----------------|
|                     |               |                |                   |                 |
|                     |               |                |                   |                 |
|                     |               |                |                   |                 |
|                     |               |                |                   |                 |

### Reflection checkpoint

- Which App Service setting is most important for reducing exposure?
- Which change would require application owner approval?

## Lab 05C - Defender for Containers

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 45 minutes                                                  |
| Delivery mode  | Optional workload track                                     |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Review container and AKS security posture using Defender for Cloud.
- Identify workload protection coverage for Kubernetes clusters, registries, images, and runtime signals.
- Document recommended hardening actions for containerized workloads.

### Scenario

A Kubernetes-based workload is running in a lab AKS environment. You will review security posture, plan coverage, and findings without making risky cluster changes.

### Before you start

- AKS cluster or containerized workload provided by the instructor.
- Defender for Cloud enabled and visible for the assigned scope.
- Network and permission requirements confirmed by the instructor.
- Optional: Azure CLI, kubectl, and container registry access only if the class lab requires CLI validation.

**Safety rule:** Do not deploy containers, change cluster policies, or modify production namespaces unless explicitly instructed. AKS resources can generate cost even when idle.

### Student tasks

40. Locate the assigned AKS cluster, container registry, or container workload in Azure.
41. Review Defender for Cloud plan coverage for Containers in Environment settings or the Coverage workbook.
42. Open Defender for Cloud recommendations filtered for Kubernetes, containers, registries, or images.
43. Identify findings related to cluster configuration, image vulnerabilities, privileged containers, network exposure, policy add-ons, or runtime protection.
44. If instructed and authorized, use Azure portal or approved CLI commands to confirm cluster/resource status. Do not run destructive commands.
45. Document three prioritized recommendations and the operational owner likely responsible for each one.

### Evidence to submit

- Screenshot of container or AKS resource overview.
- Screenshot of container-related recommendations or coverage status.
- Completed container security worksheet.

### Student worksheet

| Container / AKS finding | Affected scope | Security concern | Recommended action | Owner |
|-------------------------|----------------|------------------|--------------------|-------|
|                         |                |                  |                    |       |
|                         |                |                  |                    |       |

|  |  |  |  |  |
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### Reflection checkpoint

- Which finding suggests the largest attack surface?
- Which remediation would require coordination with platform engineering or DevOps?

## Lab 05D - Defender for AI Services and Microsoft Foundry

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 45 minutes                                                  |
| Delivery mode  | Optional workload track                                     |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Review AI workload security posture using Defender for Cloud and related Microsoft security views.
- Identify AI-specific risks such as data leakage, prompt injection, over-privileged identities, and missing guardrails.
- Document security controls for AI services, Microsoft Foundry resources, model endpoints, and workload identities.

### Scenario

An organization is adopting generative AI workloads. You must assess the security posture of AI resources and explain how Defender for Cloud, identity controls, and data governance reduce risk.

### Before you start

- Azure AI service or Microsoft Foundry lab resource assigned by the instructor, if doing live review.
- Defender for Cloud enabled and relevant plan visible for the assigned scope.
- Owner or Contributor permission only if enabling the AI protection plan is part of the authorized lab.
- Familiarity with AI service concepts, model deployments, prompts, and data leakage risks.

**Safety rule:** Never submit confidential, personal, customer, regulated, or proprietary information into AI prompts. Use only instructor-provided test prompts and sample data.

### Student tasks

46. Locate the assigned Azure AI, Azure OpenAI, or Microsoft Foundry resource, if available.
47. Review Defender for Cloud recommendations, AI workload insights, or Data & AI security dashboard views available for the assigned environment.
48. Record whether AI workload protection is enabled, disabled, unavailable, or observation-only.
49. Review identity and access configuration for the AI workload: human access, managed identities, app registrations, role assignments, and least privilege concerns where visible.
50. Review available guardrail or policy controls for model inputs, outputs, prompt handling, data access, and content safety where applicable.
51. If a demo alert or simulated case is provided, investigate the alert context and record the evidence without changing production settings.
52. Document three AI-specific risks and the controls that would reduce them.

### Evidence to submit

- Screenshot of AI resource or AI security dashboard view, with sensitive identifiers masked.

- Completed AI workload security worksheet.
- Short paragraph explaining how AI security posture differs from traditional workload posture.

### Student worksheet

| AI risk | Where observed | Potential impact | Relevant control | Recommended action |
|---------|----------------|------------------|------------------|--------------------|
|         |                |                  |                  |                    |
|         |                |                  |                  |                    |
|         |                |                  |                  |                    |
|         |                |                  |                  |                    |

### Reflection checkpoint

- Which AI risk is primarily an identity risk?
- Which AI risk requires data governance rather than only infrastructure hardening?

## Lab 06 - Threat Mapping and Remediation Planning

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 35 minutes                                                  |
| Delivery mode  | Individual or team-based                                    |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Map posture findings to practical threat scenarios and mitigation strategies.
- Prioritize remediation using risk, feasibility, business impact, and operational ownership.
- Build a student remediation plan that is ready for instructor review.

### Scenario

You have collected posture, compliance, workload protection, and AI security findings. Your task is to turn those findings into a remediation plan that a technical team or customer stakeholder can act on.

### Before you start

- Completed outputs from Labs 01 to 05.
- At least three findings from Defender for Cloud or the selected workload labs.
- Awareness of change approval and rollback expectations.

**Safety rule:** Do not perform remediation actions in live environments unless authorized. A well-documented remediation recommendation is acceptable evidence.

### Student tasks

53. Select the top three findings from your prior labs.
54. For each finding, identify the likely threat scenario: data exposure, identity compromise, public attack surface, vulnerable workload, insecure configuration, missing monitoring, or AI-specific misuse.
55. Map each finding to one or more control themes: Zero Trust, least privilege, defense-in-depth, CSPM, CWPP, identity governance, monitoring, compliance, or data protection.
56. Assign a remediation priority: P1 urgent, P2 high, P3 medium, or P4 backlog.
57. Document the remediation owner, estimated effort, change risk, rollback action, and expected evidence of completion.
58. Prepare a short remediation summary for the capstone operational review.

### Evidence to submit

- Completed remediation plan worksheet.
- Evidence screenshots for each selected finding.
- One executive summary paragraph describing the top remediation theme.

### Student worksheet

| Priority | Finding | Threat scenario | Control theme | Owner | Remediation action | Evidence of closure |
|----------|---------|-----------------|---------------|-------|--------------------|---------------------|
|          |         |                 |               |       |                    |                     |
|          |         |                 |               |       |                    |                     |
|          |         |                 |               |       |                    |                     |
|          |         |                 |               |       |                    |                     |

### Reflection checkpoint

- Which remediation gives the greatest risk reduction for the least disruption?
- Which remediation requires governance approval?

## Lab 07 - Operational Review Capstone

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 45 minutes                                                  |
| Delivery mode  | Team-based or individual presentation                       |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

### Learning objectives

- Synthesize strategy, posture, workload protection, remediation, and operational readiness findings.
- Prepare a concise operational review suitable for a security leader or cloud platform owner.
- Explain practical next steps for improving Microsoft Defender for Cloud adoption and cloud security maturity.

### Scenario

You are briefing stakeholders after a cloud security envisioning workshop. Your review must be short, evidence-based, and practical. It must explain the current posture, the most important risks, the selected workload protection gaps, and recommended next steps.

### Before you start

- Completed labs and evidence pack.
- At least three prioritized findings.
- Instructor-specified presentation format or LMS submission location.

**Safety rule:** Do not disclose sensitive identifiers in screenshots or presentations. Use generalized names unless the environment owner approves disclosure.

### Student tasks

59. Create a one-page operational review summary using the capstone worksheet.
60. Summarize the current security posture in two to three sentences.
61. List the top three risks, the affected workload or control area, and the business impact.
62. List the workload protection coverage state for the selected two workload modules.

63. Recommend immediate next steps for the next 7 days, 30 days, and 90 days.
64. Identify any blockers related to permissions, paid plans, data governance, change approval, or operational ownership.
65. Submit or present the summary as directed by your instructor.

#### Evidence to submit

- Completed capstone worksheet or one-page summary.
- Evidence pack containing the required screenshots and worksheets.
- Verbal or written explanation of the recommended next steps.

#### Student worksheet

| Section | Student summary |
|---------|-----------------|
|         |                 |
|         |                 |
|         |                 |
|         |                 |
|         |                 |
|         |                 |

#### Reflection checkpoint

- Which recommendation would you defend to an executive sponsor?
- What must be true before this environment is ready for ongoing Defender for Cloud operations?

## Lab 08 - Cleanup and Cost Control

| Field          | Details                                                     |
|----------------|-------------------------------------------------------------|
| Estimated time | 20 minutes                                                  |
| Delivery mode  | Individual or instructor-led                                |
| Student output | Evidence pack entry, completed worksheet, and blocker notes |

#### Learning objectives

- Confirm that temporary lab resources, role assignments, and paid plans are cleaned up according to scope.
- Reduce the risk of unexpected Azure cost after the course.
- Document final lab state and unresolved cleanup actions.

#### Scenario

Hands-on labs may create Azure resources or enable paid plans. Before the course closes, you must confirm what should be deleted, disabled, retained, or handed over.

#### Before you start

- Instructor confirmation of what resources were created or modified during labs.
- Permission to delete your assigned lab resources or request cleanup by the instructor.
- Awareness of the naming convention used for the lab.

**Safety rule:** Only delete resources in your assigned lab scope. Do not disable plans or delete resources that are part of a live customer environment unless explicitly instructed.

### Student tasks

66. List all resources you created or modified during the labs.
67. Confirm whether each resource should be retained, stopped, deleted, or handed over.
68. Delete or stop lab-only resources as instructed: storage accounts, App Services, AKS clusters, registries, AI resources, model deployments, or temporary resource groups.
69. Confirm whether any paid Defender plan was enabled for the lab and whether it should remain enabled or be disabled by the instructor or subscription owner.
70. Remove temporary role assignments if you were instructed to do so and have permission.
71. Record any cleanup task that could not be completed and assign an owner.

### Evidence to submit

- Completed cleanup checklist.
- Screenshot or note confirming resource deletion, stop state, or handover decision.
- List of unresolved cleanup actions and owners.

### Student worksheet

| Resource / setting | Action required | Completed? | Owner | Evidence / note |
|--------------------|-----------------|------------|-------|-----------------|
|                    |                 |            |       |                 |
|                    |                 |            |       |                 |
|                    |                 |            |       |                 |
|                    |                 |            |       |                 |
|                    |                 |            |       |                 |

### Reflection checkpoint

- Which resource or setting could create unexpected cost after the lab?
- Which cleanup action requires instructor or subscription owner approval?

## Appendix A - Student Evidence Pack Template

Use the following structure when submitting your lab evidence. Your instructor may provide a Microsoft Teams, Loop, LMS, GitHub, or SharePoint submission location.

| Evidence item   | Required content                                                                               |
|-----------------|------------------------------------------------------------------------------------------------|
| Cover note      | Name, course, lab date, assigned tenant/subscription/resource group, selected workload tracks. |
| Lab 00 evidence | Access validation screenshot and blocker log if applicable.                                    |
| Lab 01 evidence | Security goals, scope statement, and initial risk baseline.                                    |
| Lab 02 evidence | Secure Score/posture screenshot and three selected recommendations.                            |
| Lab 03 evidence | Compliance dashboard screenshot and compliance gap worksheet.                                  |
| Lab 04 evidence | Workload protection selection matrix and plan status notes.                                    |
| Lab 05 evidence | Evidence for the selected two workload labs only.                                              |
| Lab 06 evidence | Threat mapping and remediation plan.                                                           |
| Lab 07 evidence | Operational review summary.                                                                    |
| Lab 08 evidence | Cleanup checklist if resources were created or modified.                                       |



## Appendix B - Standard Lab Naming Convention

Use the naming convention supplied by the instructor. If none is provided, use the pattern below and replace <studentid> with your assigned student identifier, not personal information.

| Resource type         | Example naming pattern         |
|-----------------------|--------------------------------|
| Resource group        | rg-cese323-lab-<studentid>     |
| Storage account       | stcese323<studentid><random>   |
| App Service           | app-cese323-<studentid>        |
| AKS cluster           | aks-cese323-<studentid>        |
| Container registry    | acrce323<studentid><random>    |
| AI / Foundry resource | ai-cese323-<studentid>         |
| Budget alert          | budget-cese323-lab-<studentid> |

## Appendix C - Blocker Log Template

| Time | Lab | Blocker | Impact | Action requested | Owner | Status |
|------|-----|---------|--------|------------------|-------|--------|
|      |     |         |        |                  |       |        |
|      |     |         |        |                  |       |        |
|      |     |         |        |                  |       |        |
|      |     |         |        |                  |       |        |

## Appendix D - Student Safety Checklist

- I am working in the assigned lab tenant, subscription, and resource group.
- I have not used real customer, HR, legal, financial, regulated, or confidential data.
- I have not uploaded malware, unauthorized test files, or confidential AI prompts.
- I have not changed production resources without explicit written approval and instructor direction.
- I have recorded cost-impacting changes and cleanup actions.
- I have masked sensitive identifiers in screenshots where required.
- I have documented blockers rather than attempting unsafe workarounds.

## Appendix E - Glossary

| Term               | Meaning in this guide                                                                                                              |
|--------------------|------------------------------------------------------------------------------------------------------------------------------------|
| CSPM               | Cloud Security Posture Management. Used to identify, prioritize, and remediate configuration and posture risks.                    |
| CWPP               | Cloud Workload Protection Platform. Used to protect workloads such as servers, containers, databases, storage, and AI services.    |
| Defender for Cloud | Microsoft cloud security posture and workload protection service used throughout the labs.                                         |
| Defender XDR       | Microsoft unified security operations experience for incidents, alerts, hunting, and cross-domain security context.                |
| Secure Score       | A posture scoring mechanism used to prioritize recommendations and track improvement.                                              |
| Attack path        | A chain of exposures or weaknesses that could allow an attacker to reach a critical asset.                                         |
| Microsoft Entra ID | Microsoft identity and access platform used for users, groups, roles, MFA, Conditional Access, and workload identities.            |
| Microsoft Foundry  | Microsoft platform for building and managing AI solutions; used in this guide for AI workload security scenarios where applicable. |

## Appendix F - Reference Links

- Microsoft Learn collection for the course: <https://learn.microsoft.com/en-us/collections/grx3igtcpwee6g?source=docs&sharingId=6319874F856A7FF8>
- Azure portal: <https://portal.azure.com/>
- Microsoft Defender portal: <https://security.microsoft.com/>
- Microsoft Entra admin center: <https://entra.microsoft.com/>
- Defender for Cloud - connect Azure subscription: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/connect-azure-subscription>
- Defender for Cloud permissions: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/permissions>
- Defender for Storage support matrix: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/support-matrix-defender-for-storage>
- Defender for Containers enablement: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/defender-for-containers-enable-plan>
- Defender for AI Services onboarding: <https://learn.microsoft.com/en-us/azure/defender-for-cloud/ai-onboarding>

## Appendix G - Final Student Completion Record

| Lab                                          | Completed? | Evidence submitted? | Student note |
|----------------------------------------------|------------|---------------------|--------------|
| Lab 00                                       |            |                     |              |
| Lab 01                                       |            |                     |              |
| Lab 02                                       |            |                     |              |
| Lab 03                                       |            |                     |              |
| Lab 04                                       |            |                     |              |
| Lab 05A / 05B / 05C / 05D<br>selected tracks |            |                     |              |
| Lab 06                                       |            |                     |              |
| Lab 07                                       |            |                     |              |
| Lab 08                                       |            |                     |              |